

Charge Air Receiver Temperature Signal for LLOYD's Malfunction

Symptoms

- The OEM signal for LLOYD's sensor has malfunctioned.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

This LLOYD's sensor is connected to the OEM side (X7 connector) of the remote input/output unit. The OEM is responsible for this connection.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the OEM wiring harness.	
	STEP 1A. Check the charge air receiver temperature signal, return, and return 2 wires for an open.	
	STEP 1B. Check the charge air receiver temperature signal, return, and return 2 wires for a wire-to-wire short.	
	STEP 1C. Check the charge air receiver temperature signal wire for a short to ground.	

STEP 1. Check the OEM wiring harness.

STEP 1A. Check the charge air receiver temperature signal, return, and return 2 wires for an open.

- Conditions :**
- Open the customer interface box.
 - Disconnect the charge air receiver temperature signal, return, and return 2 wires at the remote input/output unit X7 connector.

Action	Specification/Repair	Next Step
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Check the charge air receiver temperature signal, return, and return 2 wires for an open. • Place one test lead on the charge air receiver temperature signal pin at the X7 connector. • Place the other test lead on the charge air receiver temperature signal pin at the sensor connector. • Place one test lead on the charge air receiver temperature return pin at the X7 connector. • Place the other test lead on the charge air receiver temperature return pin at the sensor connector. • Place one test lead on the charge air receiver temperature return 2 pin at the X7 connector. • Place the other test lead on the charge air receiver temperature return 2 pin at the sensor connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to the OEM service manual.	Repair complete

STEP 1B. Check the charge air receiver temperature signal, return, and return 2 wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the charge air receiver temperature signal, return, and return 2 wires at the remote input/output unit X7 connector.

Action	Specification/Repair	Next Step
Check the charge air receiver temperature signal, return, and return 2 wires for a wire-to-wire short. • Place one test lead on the charge air receiver temperature signal pin at the X7 connector. • Place the other test lead on all other pins at the X7 connector. • Place one test lead on the charge air receiver temperature return pin at the X7 connector. • Place the other test lead on all other pins at the X7 connector. • Place one test lead on the charge air receiver temperature return 2 pin at the X7 connector. • Place the other test lead on all other pins at the X7 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO	1C

STEP 1C. Check the charge air receiver temperature signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the charge air receiver temperature signal wire at the remote input/output unit X7 connector.

Action	Specification/Repair	Next Step
Check the charge air receiver temperature signal wire for a short to ground. • Place one test lead on the charge air receiver temperature signal pin at the X7 connector. • Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO Repair : Replace the charge air receiver temperature sensor. Refer to the OEM service manual or contact a Cummins® Authorized Repair Location.	Repair complete